



ORIGINAL RESEARCH

Comparative Study of Ultrasonography and Ultra-Widefield Fundus Photographs for Measurements of the Diameter of Choroidal and Retinal Tumors

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ABSTRACT

Introduction: Measurement of the largest basal dimension (LBD) of intraocular tumors is important as a prognostic parameter. To evaluate the potential value of true color ultra-widefield fundus photography for measuring tumors, we compared LBD measurements of choroidal and retinal tumors using a color ultra-widefield fundus camera with clinical estimation based on indirect ophthalmoscopy and standardized ophthalmic ultrasound.

Methods: The LBD of 148 choroidal and retinal tumors in 148 patients seen at Tongren Hospital were measured using ultra-widefield fundus photography and compared with measurements obtained using B-scan ultrasonography and clinical estimates based on indirect ophthalmoscopy.

Results: Paired *t*-tests and Bland–Altman plots reveal that measurements from ultra-widefield fundus photographic images are not statistically different from clinical estimates and ultrasound measurements. The results also showed that, although not statistically significant, when the tumor boundary was clear, the height was < 3 mm, or the tumor was pigmented, measurement from ultra-widefield fundus photography tended to be greater than those obtained by ultrasound.

Conclusions: The LBD measurement using ultra-widefield fundus photography correlated well with ultrasonography and clinical estimation and could be used as a reliable tool for measuring the LBD of choroidal and retinal tumors.

Keywords: Choroidal tumor; Retinal tumor; Largest basal diameter; True-color ultra-widefield photography

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Key Summary Points

Why carry out this study?

The largest basal dimension (LBD) plays an important role in staging, proper treatment selection, and follow-up of intraocular tumors.

The purpose of this study was to compare LBD measurements of choroidal and retinal tumors using true-color ultra-widefield fundus camera images with clinical estimates based on indirect ophthalmoscopy and measurements obtained using conventional standardized ophthalmic ultrasound.

What was learned from the study?

LBD measurements from ultra-widefield fundus photographic images are not statistically different from clinical estimates and ultrasound measurements.

Ultra-widefield fundus photography could serve as a reliable tool for measuring the LBD of choroidal and retinal tumors.

INTRODUCTION

The largest basal dimension (LBD) plays an important role in staging, proper treatment selection, and follow-up of intraocular tumors [1]. Methods for clinically estimating LBD include indirect ophthalmoscopy, B-scan ultrasonography, and fundus photography [2]. Ophthalmic ultrasound has been used by many ocular oncologists as the main tool for measuring the size of uveal melanoma. While the measurement of the maximal height, especially by A-scan ultrasonography, is accurate, it is well known that the measurement of the basal diameter by B-scan ultrasonography is far less accurate. [3]

Recently, an ultra-widefield digital fundus photography, the Clarus 500 (CLARUS 500, Carl

Zeiss Meditec AG, Jena, Germany), was introduced. It is a retinal imaging system designed to cover up to 133° of the retina in a single image. It has features of partially confocal optics and true-color imaging, which is not available with the Optos [4]. The Clarus 500 enables true-color, high-resolution, ultra-widefield imaging of the ocular fundus with minimal distortion [4]. It is especially useful in documenting intraocular tumors, especially choroidal melanoma, because in most cases the entire tumor can be photographed in one image. Special software enables the user to accurately measure the tumor diameter.

The purpose of this study was to compare the measurements of the LBD of choroidal and retinal tumors taken by a color ultra-widefield fundus camera with clinical estimations based on indirect ophthalmoscopy and with those obtained using conventional standardized ophthalmic ultrasound. The study also aimed to explore the value of true-color ultra-widefield fundus photography in measuring tumor LBDs.

METHODS

Participants were recruited in accordance with the principles of the Declaration of Helsinki. The study protocol was approved by the Medical Ethics Committee of Beijing Tongren Hospital. Written informed consent was obtained from the patients for participation in the study and for publication of the study findings.

Patients with choroidal or retinal tumors who underwent indirect ophthalmoscopy, ultra-widefield fundus imaging, and ophthalmic ultrasonography at the clinic of Beijing Tongren Hospital between 2020 and 2022 were retrospectively identified. Exclusion criteria were patients who did not have all specified imaging modalities on the same date, or who had imaging without clear visualization of all tumor borders in all modalities (poor image quality, presence of blink artifacts, and media opacities). Age, gender, and intraocular tumor diagnosis were recorded. For clinical estimation, a dilated fundus examination was performed by an ocular oncologist (WWB) to estimate the

maximal dimensions by indirect ophthalmoscopy. The transverse diameter of the optic disc is defined as 1.5 mm. By comparing with the optic disc, LBDs were determined using indirect ophthalmoscopy. Tumor shape, color (pigmented versus amelanotic), margins (well outlined versus poorly outlined), and associated subretinal fluid were all evaluated by indirect ophthalmoscopy performed by one ocular oncologist (WWB). For ophthalmic ultrasonography, an 18 MHz B-probe (MyLab 90, Esaote, Genova, Italy) was used. Two-dimensional ultrasound was performed to observe lesion location and acoustic characteristics. The LBD (using a single straight line joining the calipers at each border of the tumor) and tumor height (apical tumor surface to the inner sclera) were measured from B-scan images. Blood flow within the lesion was observed by color Doppler Flow imaging. Applying the Clarus fundus camera (CLARUS 500, Carl Zeiss Meditec AG, Jena, Germany), 200° fields montaged images of true color, red decomposition, green decomposition, and blue decomposition were obtained. The LBDs were then measured using the review software, which is freely available to users. The caliper line tool was used to draw the LBD of the lesion, and the image review software automatically calculated the measurement (Fig. 1). All fundus images were taken by the same photographer and anonymized with examination ID numbers.

Statistical analysis was performed using SPSS for Mac version 25.0 (IBM/SPSS, Chicago, Illinois, USA). Continuous variables were presented as mean $\pm$ standard deviation. Paired *t*-test analysis and Bland–Altman plots were used to compare ultrasonic measurements to ultra-widefield camera measurements and determine whether ultra-widefield camera measurements correlated well with clinical measurements. A *P*-value < 0.05 was considered statistically significant.

To assess the intraobserver and interobserver agreement of LBD measurements using ultra-widefield fundus photography, we randomly selected the image of 30 eyes of participants. For the determination of intraobserver variability, LBD was assessed twice by one examiner, and the intraclass correlation coefficient was calculated. The re-examinations took place at

intervals of 1–2 weeks. For the determination of the interobserver variability, the parameter was assessed once by both examiners, and we calculated the Bland–Altman plot.

RESULTS

The study primarily included 192 individuals (89/46.4% men) with a mean age of 44.8 ± 15.2 years (median 45.2 years; range 2.4–82.3 years). Due to the following reasons, ultra-widefield fundus photography could not clearly or completely show the tumor and the measurement of the tumor diameter was not available in 44 patients: the tumor was too big or too close to the periphery (23/11.9% patients), retinal detachment (2/1.0% patients), severe vitreous opacity (9/4.7% patients), and very unclear boundary (10/5.2% patients). Eventually, 148 tumors in 148 eyes were included in the present study. The demographic and clinical features are presented in Table 1. The number of photos used for measurement is 148 for true-color and red photos, 144 for green photos (4 being excluded due to unrecognizable boundary), and 126 for blue photos (22 being excluded due to unrecognizable boundary).

The mean LBD was 9.9 ± 5.6 mm (median 8.3 mm; range 1.5–27.0 mm) when clinically estimated based on indirect ophthalmoscopy, 9.5 ± 3.5 mm (median 9.8 mm; range 2.3–17.4 mm) when measured by ultrasonography, 9.6 ± 4.6 mm (median 8.7 mm; range 1.8–21.2 mm) when assessed by ultra-widefield fundus true-color photos, 9.5 ± 4.6 mm (median 8.5 mm; range 1.8–21.2 mm) for ultra-widefield fundus red photos, 9.6 ± 4.6 mm (median 8.7 mm; range 1.8–21.2 mm) for ultra-widefield fundus green photos, and 9.4 ± 4.6 mm (median 8.6 mm; range 1.8–21.1 mm) for ultra-widefield fundus blue photos.

Paired *t*-tests reveal that measurements from ultra-widefield fundus photographic true-color images are not statistically different from clinical estimation and ultrasound measurements. The Bland–Altman plot conducted to test agreement of ultra-widefield fundus photographic true-color images measurements and

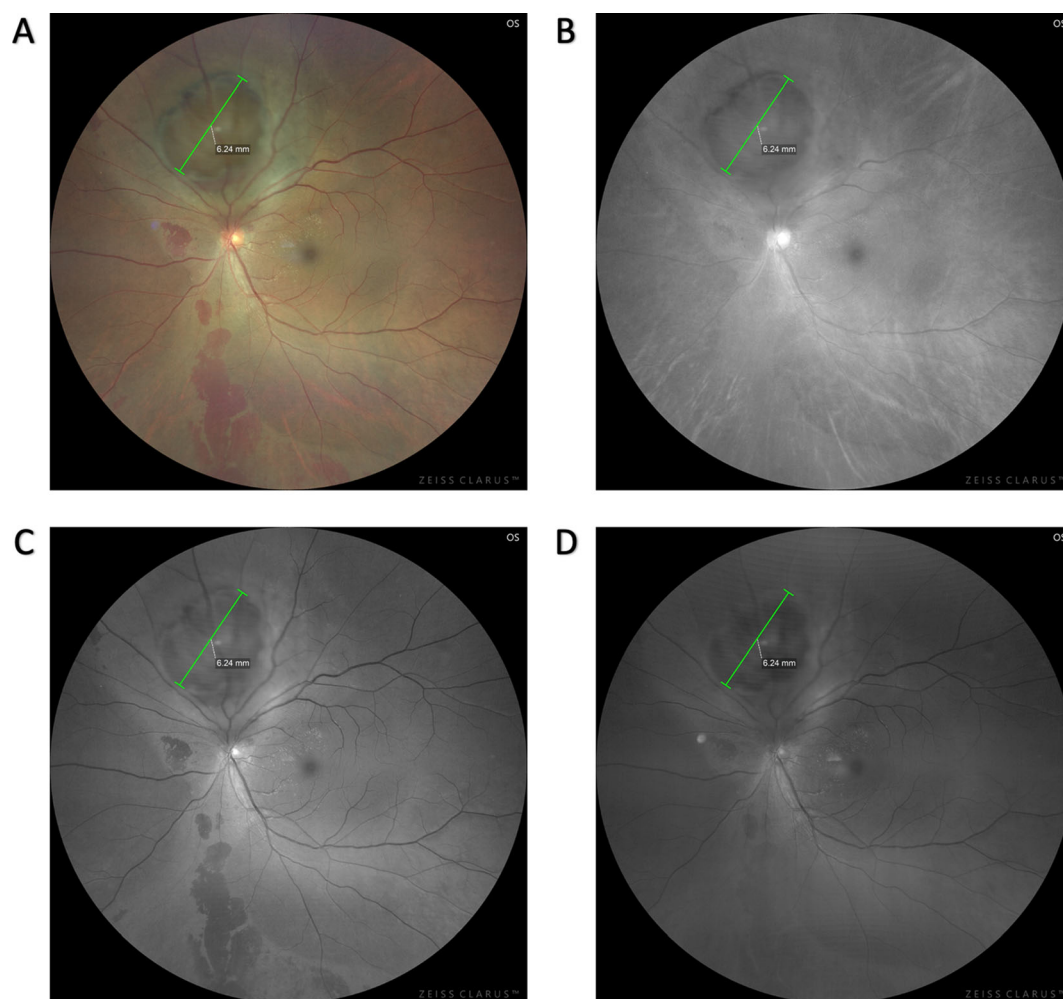


Fig. 1 Measurement of the largest basal dimension by using caliper line tool of the image review software and images of true color, red decomposition, green decomposition, and blue decomposition (A–D)

ultrasonography measurements demonstrated normal distributions; the mean difference was around zero (-0.12) and 95% of the measurements fell within 1.96 SD of the mean. Furthermore, the difference in variability between the two measurements (ultra-widefield fundus photographic true-color images and ultrasonography) was nonsignificant ($P = 0.662$) (Fig. 2). The Bland–Altman plots for ultra-widefield fundus photographic true-color images measurements and indirect ophthalmoscope measurements demonstrated normal distributions, and 95% of the measurements fell within 1.96 SD of the mean. The difference in variability between the two measurements was nonsignificant ($P = 0.218$) (Fig. 3).

Measurements with ultra-widefield red images ($P = 0.049$), green images ($P = 0.035$), and blue images ($P = 0.015$) were significantly smaller than clinical estimations (Table 2).

Since the tumor's boundary, height, and color have a great influence on the fundus photographic measurement [1, 3], we grouped them according to whether the tumor boundary was clear, whether the tumor height was greater than 3 mm or less than 3 mm, or whether the tumor was pigmented or amelanotic. We then compared the difference between ultra wide-field fundus photographic measurements and clinical evaluations and ultrasound measurements. The results showed that, although not statistically significant, when the tumor

Table 1 Participants' demographics and tumor features

Gender (female/male)	81/67
Mean age (years)	44.3 ± 14.8
Diagnosis	
Melanoma	97 (65.5%)
Melanocytoma	7 (4.7%)
Hemangioma	14 (14.8%)
Osteoma	6 (4.1%)
Retinoblastoma	3 (2.0%)
Metastasis	7 (4.7%)
Nevus	2 (1.4%)
Other occupying lesions	12 (8.1%)
Tumor shape	
Dome	111 (75.0%)
Other	37 (25.0%)
Tumor height	
< 3 mm	62 (41.9%)
≥ 3 mm	86 (58.1%)
Tumor margins	
Well outlined	92 (62.2%)
Poorly outlined	56 (37.8%)
Tumor color	
Pigmented	102 (68.9%)
Amelanotic	46 (31.1%)
Subretinal fluid	
Present	51 (34.5)
Absent	97 (65.5%)

boundary was clear, the height was < 3 mm, or the tumor was pigmented, the measurement value from ultra-widefield fundus photography was greater than that measured by ultrasound (Table 3).

The intraclass correlation coefficient for the measurements of LBD using ultra-widefield fundus photography was 0.878 ($P < 0.001$), indicating a high intraobserver agreement.

When the interobserver variability was assessed, the Bland–Altman plot showed that 6.7% (2/30) of the points of the LBD measurements were located outside the 95% limits of agreement. This revealed a high interobserver agreement.

DISCUSSION

The LBD of intraocular tumors is a very important parameter for evaluating prognosis [5–7], and an accurate measurement is even more important for practical use, such as planning the size and shape of the radioactive applicator for use in brachytherapy [8, 9]. Ultrasonography has been and continues to be the most commonly used method for measuring the LBD of intraocular tumors [10]. Although A-scan ultrasonography is highly accurate in measuring the maximal thickness of the tumor, the accuracy of measuring the LBD is quite problematic because tumors must be greater than 0.4 mm in height to be identified by ultrasound. Additionally, ultrasonic measurements depend on placing the cursor on the ultrasound screen at relatively low magnifications, which could also cause some inaccuracies [3]. Ultrasonography can be unpredictable and less accurate in follow-up because of the difficulty in repeating the exact same scanning location and plane [11]. According to the Collaborative Ocular Melanoma Study (COMS), in 644 eyes, ultrasonographic and histopathologic measurements of tumor height agreed within ± 2 mm in 90% of the tumors. However, the agreement of the measured LBD between the two measurement methods was poor, with only 58% of the 664 eyes measured within ± 2 mm [12]. Peyster et al. have also suggested that ultrasonographic measurements of the LBD are unreliable, because the exact ultrasonographic planes are difficult to reproduce [13].

Large choroidal lesions or metastases lesions with serous detachments of retina are typically difficult for traditional fundus cameras to capture. Advances in resolution and stereopsis, expanded field of view, and development of analysis tools in digital fundus photographic systems have expanded the use of the instrument into new areas of research [14]. The ultra-

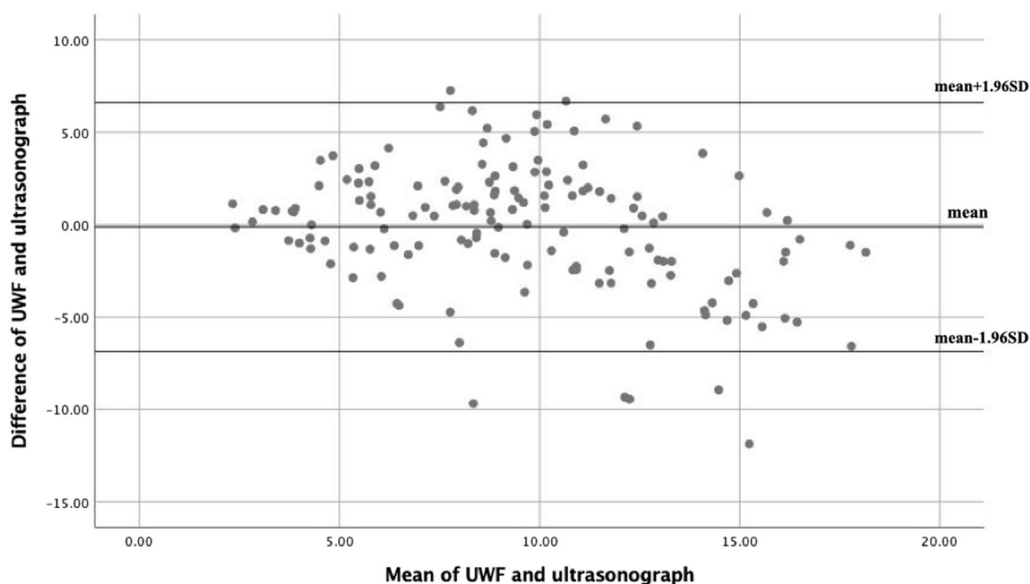


Fig. 2 Bland–Altman plot for ultra-widefield (UWF) fundus true-color photography measurements and ultrasonography measurements

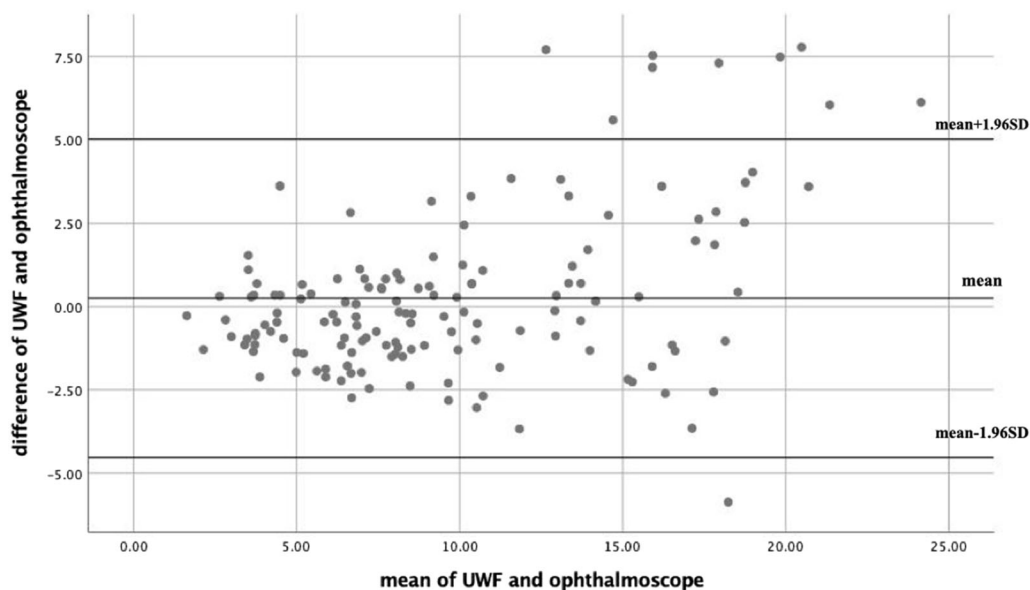


Fig. 3 Bland–Altman plot for ultra-widefield (UWF) fundus true-color photography measurements and indirect ophthalmoscope measurements

widefield fundus camera is well suited for tele-medicine in under-served areas or for minimizing the economic burden of ultrasonography examinations. With montaged photos, the entire extent of a tumor can be imaged, and analytical tools can directly measure the LBD.

In theory, ultra-widefield photographic images allow for more accurate measurements of tumor size. Another benefit of fundus photogrammetry is the ability to accurately measure the distance between the tumor margin and important intraocular structures such as the optic disc and

Table 2 Paired *t*-test comparison results of different measurement methods

Measurement technique	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)
Clinical estimation	–	–
Ultrasound	0.29	–
Widefield color	0.19	0.66
Widefield red	0.049	0.98
Widefield green	0.035	0.93
Widefield blue	0.015	0.76

macula. This measurement can more accurately predict the likelihood of radiation maculopathy or optic neuropathy. This study suggests that tumor measurements in ultra-widefield fundus photographic true-color images correlate significantly with both clinical assessments and ultrasound measurements. Ultra-widefield fundus photographic true-color image tumor measurements were larger than ultrasonographic measurements when tumor boundaries were well defined, tumor height was < 3 mm, or tumors were pigmented, although not statistically significant. This is consistent with previous research results. Kim et al. compared digital fundus photography with ultrasonography in measuring the basal dimension from eyes with choroidal melanoma and found that the mean basal dimension by these two methods was within 1.1 mm in 52% of eyes and 2.2 mm in 95% of eyes [14]. They found that the digital photographic analysis estimated the LBD to be greater than the ultrasonographic measurements in most of the eyes [14]. Pe'er et al. also reported larger measurements with fundus photographs than ultrasound evaluations [3]. There may be two reasons for this. First, pigmentation of the tumor is easy to recognize and elevation in ultrasound can be recognized only above a certain size; it is assumed that measurement by an ultra-widefield fundus camera might be more accurate for measuring the LBD of clearly outlined tumors [3]. Second, as

echographic method is less reliable in areas of the lesion under 1 mm in height, fundus photographs reliably capture these relatively flat portions of the tumor, resulting in a larger LBD than ultrasound. Moreover, ultrasound can display the three-dimensional structure of the lesion, which may vary greatly due to factors such as laser, transpupillary thermotherapy, brachytherapy, surgery, tamponades used (oil), or a combination of therapies.

In the present study, we found that true-color and red photos show tumor boundaries better than green and blue photos. We found that tumors that did not show borders in green-light and blue-light photographs were all choroidal tumors. This may be because red light has stronger penetrating power, which can better show the details of choroidal lesions and help visualize choroidal lesions. The penetrating power of green light and blue light successively decreases, and they are used to display the retina and superficial retina (such as retinal nerve fiber layer defects and macular membrane), respectively.

The methodologies employed for assessing LBD in the current investigation possess inherent limitations. Clinical estimation reliant upon indirect ophthalmoscopy represents a subjective approach, necessitating the expertise of a proficient and experienced ophthalmologist to ensure accurate evaluations, while also presenting challenges when attempting to compare results across different practitioners. Ultrasonography, on the other hand, encounters challenges in accurately measuring LBD due to the requirement for tumors to exceed a height threshold of 0.4 mm for identification. The use of ultrasonography during follow-up evaluations introduces additional complexities, as replicating identical scanning locations and plans becomes arduous, thereby undermining the predictability and precision of subsequent measurements. The measurement of ultra-widefield fundus photography also has certain limitations. Compared with ultrasound, fundus photographs present two-dimensional images. When vitreous opacity, subretinal fluid, or retinal detachment obscures the tumor, or the tumor is too large or is located in the peripheral retina and close to the ciliary body, it is difficult

Table 3 After grouping according to tumor border, height, and pigment, the results of paired *t*-test comparisons of different measurement methods

Measurement technique	Tumors with well-outlined boundary			Tumors with poorly outlined boundary		
	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)
Clinical estimation	10.2 $\pm$ 5.8	–	–	9.5 $\pm$ 5.3	–	–
Ultrasound	9.6 $\pm$ 3.5	0.20	–	9.4 $\pm$ 3.7	0.90	–
Widefield color	9.8 $\pm$ 4.6	0.13	0.55	9.4 $\pm$ 4.6	0.88	0.99
Widefield red	9.7 $\pm$ 4.6	0.06	0.74	9.2 $\pm$ 4.5	0.45	0.70
Widefield green	9.66 $\pm$ 4.6	0.048	0.80	9.5 $\pm$ 4.6	0.38	0.89
Widefield blue	9.4 $\pm$ 4.5	0.055	0.67	9.3 $\pm$ 4.7	0.14	0.98
Measurement technique	Tumor height < 3 mm			Tumor height $\geq$ 3 mm		
	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)
Clinical estimation	6.7 $\pm$ 3.8	–	–	12.2 $\pm$ 5.6	–	–
Ultrasound	6.7 $\pm$ 2.7	0.96	–	11.6 $\pm$ 2.5	0.22	–
Widefield color	7.0 $\pm$ 3.4	0.31	0.46	11.6 $\pm$ 4.4	0.059	0.97
Widefield red	6.8 $\pm$ 3.3	0.79	0.72	11.5 $\pm$ 4.4	0.021	0.80
Widefield green	6.8 $\pm$ 3.4	0.89	0.58	11.5 $\pm$ 4.4	0.024	0.81
Widefield blue	6.8 $\pm$ 3.5	0.58	0.45	11.2 $\pm$ 4.4	0.015	0.40
Measurement technique	Pigmented tumors			Amelanotic tumors		
	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)	Mean $\pm$ standard deviation, mm	Clinical estimation <i>P</i> -value (paired <i>t</i> -test)	Ultrasound <i>P</i> -value (paired <i>t</i> -test)
Clinical estimation	10.2 $\pm$ 5.8	–	–	9.3 $\pm$ 5.2	–	–
Ultrasound	9.6 $\pm$ 3.6	0.12	–	9.4 $\pm$ 3.5	0.85	–

Table 3 continued

Measurement technique	Pigmented tumors		Ameletotic tumors		
	Mean ± standard deviation, mm	Clinical estimation P-value (paired <i>t</i> -test)	Mean ± standard deviation, mm	Clinical estimation P-value (paired <i>t</i> -test)	Ultrasound P-value (paired <i>t</i> -test)
Widefield color	9.7 ± 4.6	0.054	9.5 ± 4.6	0.49	0.90
Widefield red	0.6 ± 4.6	0.027	9.2 ± 4.5	0.93	0.79
Widefield green	9.7 ± 4.6	0.02	9.3 ± 4.7	0.86	0.96
Widefield blue	9.5 ± 4.6	0.016	9.2 ± 4.6	0.48	0.83

to obtain accurate measurement. In addition, the images were reviewed in the sequence of color, red, blue, and green, which may create a bias in the image analysis and also allows the examiner to review the color photo for reference. Lastly, the measurement of ultra-widefield fundus photography does not allow the the repeated assessment of the treatment response in the same spot in follow-up, but it is possible to relocate the lesion using the scale tool.

Potential limitations of our study include the following issues. Heterogeneity in the sample, comprising diverse intraocular tumor subtypes, may introduce systematic error due to underlying histopathological differences impacting measurement across modalities. Exclusion of a sizable proportion of patients (44/192) for reasons encompassing peripheral tumor location, retinal detachment, and vitreous opacity may introduce selection bias by favoring the inclusion of lesions more amenable to measurement. This nonrandom loss of potentially relevant cases risks overestimating the agreement between techniques by preferentially retaining those more likely to yield concordant results. Additional limitations include the cross-sectional design, which precludes evaluating changes in agreement over time.

CONCLUSION

The LBD measurement obtained using ultra-widefield fundus photography correlated well with those from ultrasonography and clinical estimation. While ultrasonography provided more accurate measurements for tumors with opaque media, retinal detachment, or peripheral retinal location, ultra-widefield fundus photography can serve as a reliable tool for measuring the LBD of choroidal and retinal tumors.

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Data Availability. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Ethical Approval. This study was conducted according to the tenets of the Declaration of Helsinki. The Medical Ethics Committee of the Beijing Tongren Hospital approved the study protocol. Written informed consent was obtained from every participant after a full explanation of the nature and possible consequences of the study.

Conflict of Interest. Qiong Yang, Qian Wang, Jing Yan Yang, and Wen Bin Wei declare that they have no competing interests.

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